

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

75% 48

Code of Conduct:

I S. Santhiya (19251136) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. 

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context(2)	Shows good understanding with minor gaps(1.5)	Partial understanding; misses key aspects(1)	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved(2)	Identifies most key issues with minor omissions(1.5)	Identifies limited issues; some irrelevant points(1)	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario(2.5)	Applies concepts with minor errors(2)	Limited or partially correct application(1.5)	Unable to apply relevant concepts
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Clarity of Response	15%	Response is clear, structured, and logically presented(1.5)	Generally clear with minor issues(1)	Somewhat unclear or poorly structured(0.5)	Disorganized and difficult to understand

	understanding of scenario (20%)	identification of key issues (20%)	Application of concepts (25%)	Decision making (20%)	clarity of response (15%)	
Q1	1.5	1.5	2.5	2	1	8.5
Q2	1	2	2	2	1	8
Q3	1.5	1.5	1.5	1.5	1.5	7.5
Q4	2	1	1	1	1.5	6.5
Q5	2	1	1.5	1.5	1	7

37.5

SIMATS ENGINEERING

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Signature of the student:

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

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As the network engineer, design the subnet allocation by performing the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid host IP address range for each subnet.**
- 3. Calculate the number of usable host addresses available in each subnet.**
- 4. Determine the remaining unused IP address range within the given network block.**

Answers:

Given

Network: 192.168.100.0/24

Departments:

- Computing → /25
- Electronics → /26
- Mechanical → /27

Step 1: School of Computing — /25

Subnet mask: /25 = 255.255.255.128

Block size: $256 - 128 = 128$

So, the first subnet is: 192.168.100.0 – 192.168.100.127

Therefore:

- Network Address: 192.168.100.0
- First Host: 192.168.100.1
- Last Host: 192.168.100.126
- Broadcast Address: 192.168.100.127

Usable hosts:

$$2^7 - 2 = 128 - 2 = 126$$

Step 2: School of Electronics — /26

Next available address: 192.168.100.128

Subnet mask: /26 = 255.255.255.192

Block size: $256 - 192 = 64$

So:

192.168.100.128 – 192.168.100.191

Therefore:

- Network Address: 192.168.100.128
- First Host: 192.168.100.129
- Last Host: 192.168.100.190
- Broadcast Address: 192.168.100.191

Usable hosts:

$$2^6 - 2 = 64 - 2 = 62$$

Step 3: School of Mechanical Engineering — /27

Next available address:

192.168.100.192

Subnet mask:

/27 = 255.255.255.224

Block size:

$$256 - 224 = 32$$

So:

192.168.100.192 – 192.168.100.223

Therefore:

- **Network Address:** 192.168.100.192
- **First Host:** 192.168.100.193
- **Last Host:** 192.168.100.222
- **Broadcast Address:** 192.168.100.223

Usable hosts:

$$2^5 - 2 = 32 - 2 = 30$$

Step 4: Remaining Unused IP Range

Original /24 network:

192.168.100.0 – 192.168.100.255

Used up to:

192.168.100.223

Therefore, remaining unused range is:

192.168.100.224 – 192.168.100.255

Department	Subnet	Network Address	Valid host range	Broad cast	Usable host
Computing	/25	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
Electronics	/26	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
Mechanical	/27	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

Total usable hosts:

126 + 62 + 30 = 218

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid range of host IP addresses.**
- 3. Calculate the number of usable hosts in each subnet.**
- 4. Identify the unused IP address range remaining within the allocated IP block.**

Answers:

Given

Network: 10.10.0.0/16

- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

Step 1: Software Development – /24

Theory:

The Software Development department gets the largest subnet because it may have the highest number of employees. A /24 subnet provides 254 usable host addresses.

Subnet mask: 255.255.255.0

Block size = 256

Network: 10.10.0.0

Host range: 10.10.0.1 – 10.10.0.254

Broadcast: 10.10.0.255

Usable hosts: $256 - 2 = 254$

Step 2: Quality Assurance – /25

Theory:

Quality Assurance requires fewer addresses than Software Development, so a smaller /25 subnet is sufficient. It provides 126 usable host addresses.

Next address: 10.10.1.0

Subnet mask: 255.255.255.128

Block size:

$$256 - 128 = 128$$

Network: 10.10.1.0

Host range: 10.10.1.1 – 10.10.1.126

Broadcast: 10.10.1.127

Usable hosts: $128 - 2 = 126$

Step 3: Human Resources – /26

Next address: 10.10.1.128

Subnet mask: 255.255.255.192

Block size:

$$256 - 192 = 64$$

Network: 10.10.1.128

Host range: 10.10.1.129 – 10.10.1.190

Broadcast: 10.10.1.191

Usable hosts: $64 - 2 = 62$

Step 4: Executive Management – /27

Theory:

Executive Management is a smaller department, so it requires fewer IP addresses. A /27 subnet provides 30 usable host addresses.

Next address: 10.10.1.192

Subnet mask: 255.255.255.224

Block size:

$$256 - 224 = 32$$

Network: 10.10.1.192

Host range: 10.10.1.193 – 10.10.1.222

Broadcast: 10.10.1.223

Usable hosts: $32 - 2 = 30$

Step 5: Unused IP Range

Theory:

The unused IP addresses are kept available for future departments, additional computers, servers, printers, or network expansion. This makes the network scalable without changing the original /16 address block.

Allocated block:

10.10.0.0 – 10.10.255.255

Used until:

10.10.1.223

So the unused range is:

10.10.1.224 – 10.10.255.255

Department	Network	Host range	Broad Cast	Usable Hosts
Software development	10.10.0.0	.1 – .254	.255	254
Quality assurance	10.10.1.0	.1 – .126	.127	126
Human resources	10.10.1.128	.129 – .190	.191	62
Execution management	10.10.1.192	.193 – .222	.223	30

Unused IP range: 10.10.1.224 – 10.10.255.255

Conclusion

Subnetting allows the company to divide the /16 network according to departmental requirements. Larger departments receive larger subnets, while smaller departments receive smaller subnets. This reduces IP address wastage and makes network management easier.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.**
- 2. Expand the compressed address back into its full hexadecimal representation.**
- 3. Identify the network prefix using the /64 prefix length.**
- 4. Calculate the total number of host addresses supported within the subnet.**
- 5. the total number of possible host addresses available within this network.**

Answers:

Theory of IPv6 Addressing

IPv6 (Internet Protocol version 6) is the newer version of the Internet Protocol designed to replace IPv4. IPv6 uses 128-bit addresses, which provides a very large number of unique IP addresses.

An IPv6 address is written using eight groups of four hexadecimal digits, separated by colons (:).

Example:

2001:0db8:abcd:0000:0000:0000:0000:0000

Important IPv6 Concepts

128-bit Address

An IPv6 address contains 128 bits divided into 8 groups. Each group contains 16 bits.

Hexadecimal Representation

IPv6 uses hexadecimal values from 0–9 and A–F. This makes the 128-bit address shorter and easier to represent.

Leading Zero Removal

Leading zeros in any group can be removed.

Example:

0db8 → db8

0000 → 0

Zero Compression

One continuous sequence of zero groups can be replaced by ::.

Example:

2001:db8: abcd: 0:0:0: 0:0

becomes:

2001:db8:abcd::

can represent multiple consecutive zero groups, but it should normally be used only once in an IPv6 address.

Prefix Length

The /64 indicates that the first 64 bits are used for the network portion and the remaining 64 bits are available for interface/host addressing.

Given

IPv6 Network:

2001:0db8:abcd:0000:0000:0000: 0000:0000/64

Step 1: Compress the IPv6 Address

Remove leading zeros:

2001:db8:abcd:0:0:0:0:0

Replace continuous zeros with ::.

Compressed form:

2001:db8:abcd::/64

Step 2: Expand the Address

represents five groups of 0000.

So:

2001:db8:abcd::

Expanded form:

2001:0db8: abcd:0000:0000:0000:0000:0000

Step 3: Find Network Prefix

Prefix = /64

First 4 groups are the network portion:

2001:0db8:abcd:0000

Network prefix:

2001:db8:abc

d::/64

Step 4: Calculate Host Bits

IPv6 has 128 bits.

Network bits = 64

Host bits:

$$128 - 64 = 64 \text{ bits}$$

Step 5: Total Host Addresses

Formula:

$$2^{64} = 18,446,744,073,709,551,616$$

IPv6 Does Not Use Broadcast

Unlike IPv4, IPv6 does not use broadcast communication. Instead, IPv6 uses multicast and anycast mechanisms for communication with multiple devices.

This helps make IPv6 communication more efficient.

Types of IPv6 Addresses

Common IPv6 address types include:

- **Unicast** – identifies a single interface.
- **Multicast** – identifies multiple interfaces.
- **Anycast** – identifies a group of interfaces, with traffic delivered to the nearest one.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

- 1. The destination subnet to which the packet belongs.**
- 2. The matching routing table entry.**
- 3. The next-hop network/interface used to forward the packet.**
- 4. The default route that should be used if the destination does not match any routing table entry.**

Answers:

Given

Routing table:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP:

192.168.2.45

Step 1: Find the Destination Subnet

192.168.2.45 starts with 192.168.2.

So it belongs to:

192.168.2.0/24

Step 2: Find Matching Routing Entry

Routing table contains:

192.168.2.0/24

Therefore, this is the matching entry.

Step 3: Find Next-Hop Network/Interface

The router forwards the packet through the interface connected to:

192.168.2.0/24

So:

Next-hop network = 192.168.2.0/24

Destination host = 192.168.2.45

The exact interface name or next-hop IP is not given in the question, so it cannot be specified.

Step 4: Default Route

If the destination does not match any entry, the router uses:

0.0.0.0/0

This is called the default route.

Item	Answer
Destination IP	192.168.2.45
Destination subnet	192.168.2.0/24
Matching entry	192.168.2.0/24
Next hop network	192.168.2.0/24
Default route	0.0.0.0/0

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.**
- 2. Identify the valid host IP address range.**
- 3. Suggest suitable IP addresses for three computers in each LAN.**
- 4. Identify the default gateway that hosts in each LAN should use for communication.**

Answers:

Given

- Administration LAN → 192.168.10.0/26
- Finance LAN → 192.168.10.64/26

For /26:

Block size = 64

Usable hosts = $64 - 2 = 62$

Step 1: Administration LAN

Range:

192.168.10.0 – 192.168.10.63

- Network Address: 192.168.10.0
- Host Range: 192.168.10.1 – 192.168.10.62
- Broadcast Address: 192.168.10.63
- Gateway: 192.168.10.1

Three suitable computers:

- PC1 → 192.168.10.2
- PC2 → 192.168.10.3
- PC3 → 192.168.10.4

Default Gateway: 192.168.10.1

Step 2: Finance LAN

Range:

192.168.10.64 – 192.168.10.127

- Network Address: 192.168.10.64
- Host Range: 192.168.10.65 – 192.168.10.126
- Broadcast Address: 192.168.10.127
- Gateway: 192.168.10.65

Three suitable computers:

- PC1 → 192.168.10.66
- PC2 → 192.168.10.67
- PC3 → 192.168.10.68

Default Gateway: 192.168.10.65

Step 3: Find Network Address

Given:

192.168.10.0/26

The starting address is:

Network Address = 192.168.10.0

Step 4: Find Broadcast Address

The next subnet starts at:

192.168.10.64

Therefore, one address before it is the broadcast address:

Broadcast Address = 192.168.10.63

Step 5: Find Valid Host Range

Network address cannot be used by a PC.

Broadcast address cannot be used by a PC.

Therefore:

First Host = 192.168.10.1

Last Host = 192.168.10.62

So, valid host range is:

192.168.10.1 – 192.168.10.62

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